Topic 7: Design (part 1)

SCSM2113 Human-Computer Interaction Fundamentals

Faculty of Computing
Universiti Teknologi Malaysia

1

01 METAPHORS AND ANALOGIES

02 INFORMATION ARCHITECTURE

2

METAPHORS AND ANALOGIES

3

3

Interface Metaphors

- Interface designed to be similar to a physical entity but also has own properties
  - e.g, desktop metaphor, web portals, search engine
- Can be based on activity, object, or a combination of both
- Exploit user's familiar knowledge, helping them to understand 'the unfamiliar'
- Conjures up the essence of the unfamiliar activity, enabling users to leverage this to understand more aspects of the unfamiliar functionality

4

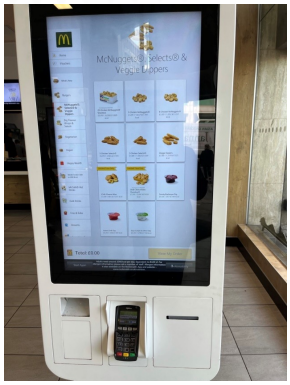
4

Examples of Interface Metaphors

- Conceptualising what users are doing
  - e.g, surfing the Web
- A conceptual model instantiated at the interface
  - e.g., the desktop metaphor
- Visualising an operation
  - e.g., an icon of a shopping cart into which the user places items

The Card Metaphor

- The card is a very popular metaphor. Why?
  - It has familiar form factor
  - It can easily be flicked through, sorted, and themed
  - Cards have strong associations, providing an intuitive way of organising content that is "card sized"
  - Structures content into meaningful chunks



A fast food display

Benefits of Interface Metaphors

- Makes learning new systems easier
- Familiar artefact or action helps people understand the underlying conceptual model
- Can be innovative and enable apps to be made accessible to a greater diversity of people

7

Problems with Interface Metaphors

- Break conventional and cultural rules
  - e.g, recycle bin placed on desktop
- Can constrain designers in the way that they conceptualize a problem space
- Can conflict with design principles
- Designers can inadvertently use bad existing designs and transfer the bad parts over
- Limits designers' imagination in coming up with new conceptual models
- Often don't scale very well, especially when trying to describe how complex systems work (Cooper, 2020)

8



Activity



- Describe the components of the conceptual model underlying most shopping websites, for example:
  - Shopping cart
  - Proceeding to check-out
  - 1-click
  - Gift wrapping
  - Cash register

INFORMATION ARCHITECTURE

11

11

by sorting out things like this:

CLASSIFICATION and HIERARCHY

LABELS and TAGGING

NAVIGATION and WAYFINDING

SEARCH

INFORMATION ARCHITECTURE

(IA for short)

connects PEOPLE to the CONTENT they're looking for

The 'content' people are looking for could be text, images, videos, whole documents...

...or even conversations...

...and other people.

...and links content to other content to make it more usable and easy to find.

IA also helps the people who create and maintain content do their job easier...

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12

12

What is Information Architecture?

- Focuses on organising, structuring, and labeling content in an effective and sustainable way. How data is presented to users
- The goal is to help users find information and complete tasks by understanding how the pieces fit together to create the larger picture, how items relate to each other within the system
- Main components of IA:
  - Organisation Schemes and Structures: How you categorise and structure information
  - Labeling Systems: How you represent information
  - Navigation Systems: How users browse or move through information
  - Search Systems: How users look for information

13

13

How To?

- Defining taxonomies
  - bring together, like terms, so that they are sensibly classified
- To get these taxonomies right:
  - User research – structured interview to try and clarify how a user might approach a given data set
  - Card sorting technique – test how logical data sets that are already defined feel for the user to see if the users group data in the same way we (designers) did
- Sorting process include:
  - Terminology (what people call things)
  - Relationships (proximity, similarity)
  - Categories (groups and their names)



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14

14

Activity



- You are responsible for the IA of UTM Student Portal. From the recent gathered survey, most lecturers found the portal to be difficult at times, as they have to carefully look for things, and it was not so easy to understand some of the labeling
- Based on the created groups, try to sort the items for portal. You may make up your own groups names if you find them necessary.
- TWO kinds of analysis:
  - An items by groups chart shows what the groups were called and what was in each
  - An item by item chart shows which items were grouped together most often
- Online card sorting tool (example): https://www.optimalworkshop.com/